

	Work Instruction	Doc. Revision	00
	AXI V810 Panasonic Ethercat Motion Driver Assembly R&R WI	Effective Date	13 Sep 2023

This procedure is applicable to below models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL			
V810 / V810i S2EX		√	√
V810 / V810i XLT		√	√
V810i XLL		√	√
V810i XLM		√	√
V810i S3			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	13 Sep 2023	Initial Release	Tan Chia Chun

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Panasonic Ethercat Motion Driver Assembly Removal

1. Switch off the SSCA.

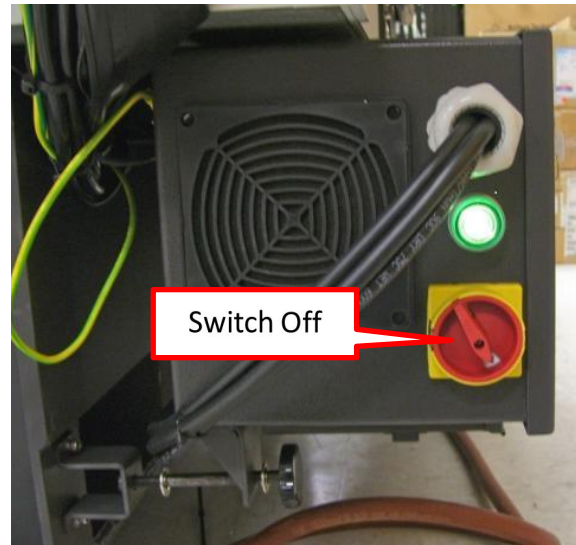


Figure 1 Turn Off SSCA

2. Identify the Motion Driver to be removed.

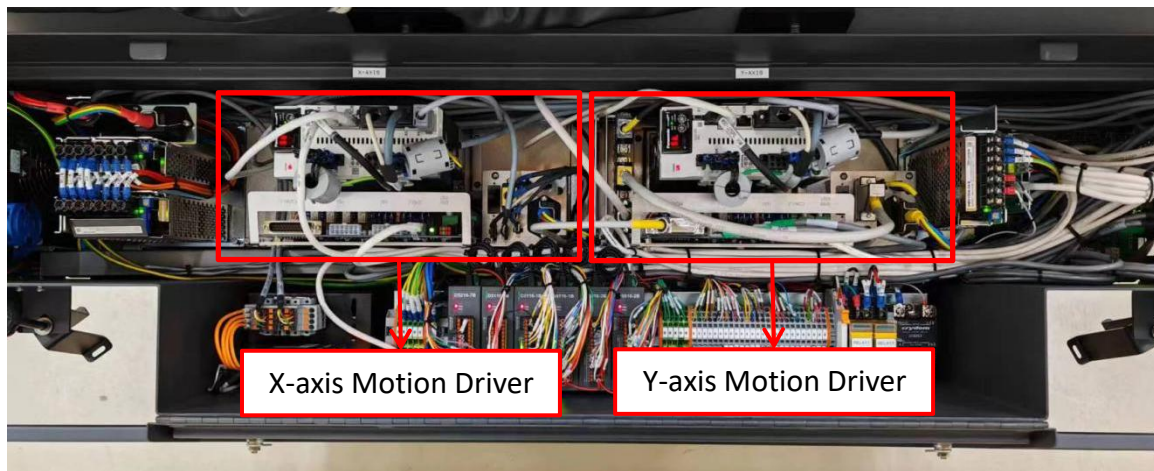


Figure 2 X & Y Axis Panasonic Ethercat Motion Driver

3. Before removing the motion driver, ensure each cable is labeled accordingly to ease the upgrade process.

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X-axis Panasonic Ethercat Motion Driver

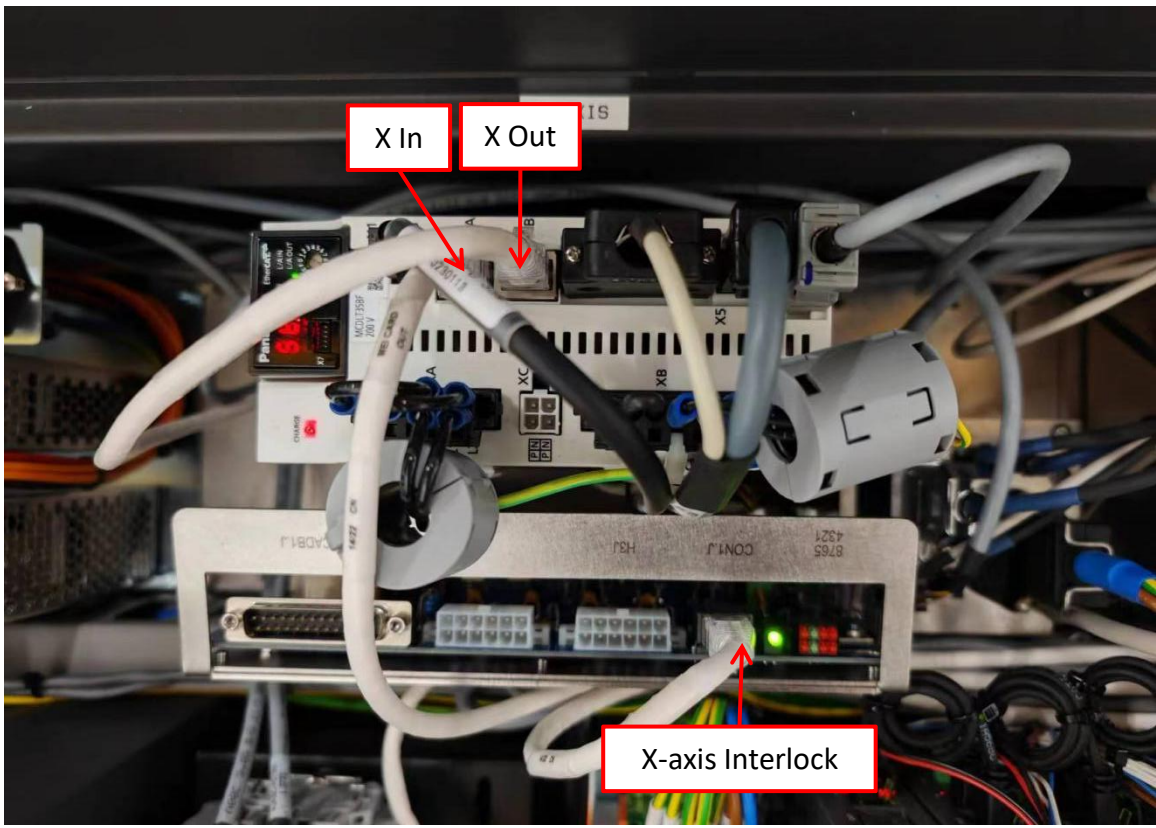


Figure 3 X-axis Panasonic Ethercat Motion Driver Cable Labeling (1)

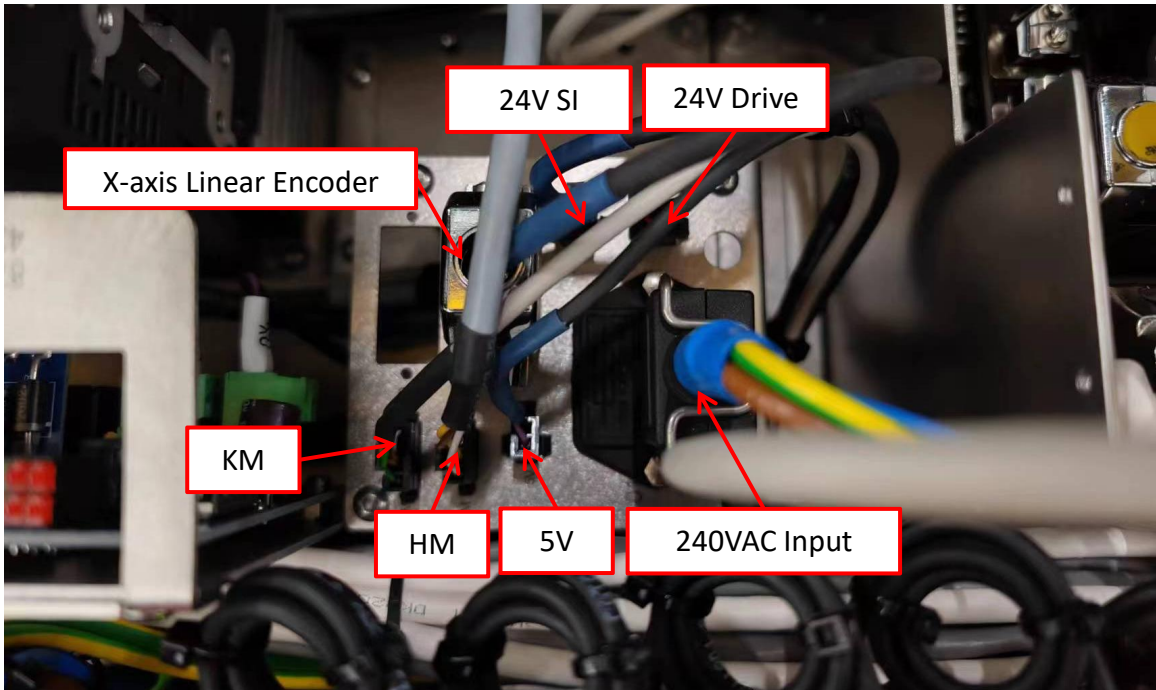


Figure 4 X-axis Panasonic Ethercat Motion Driver Cable Labeling (2)

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Y-axis Panasonic Ethercat Motion Driver

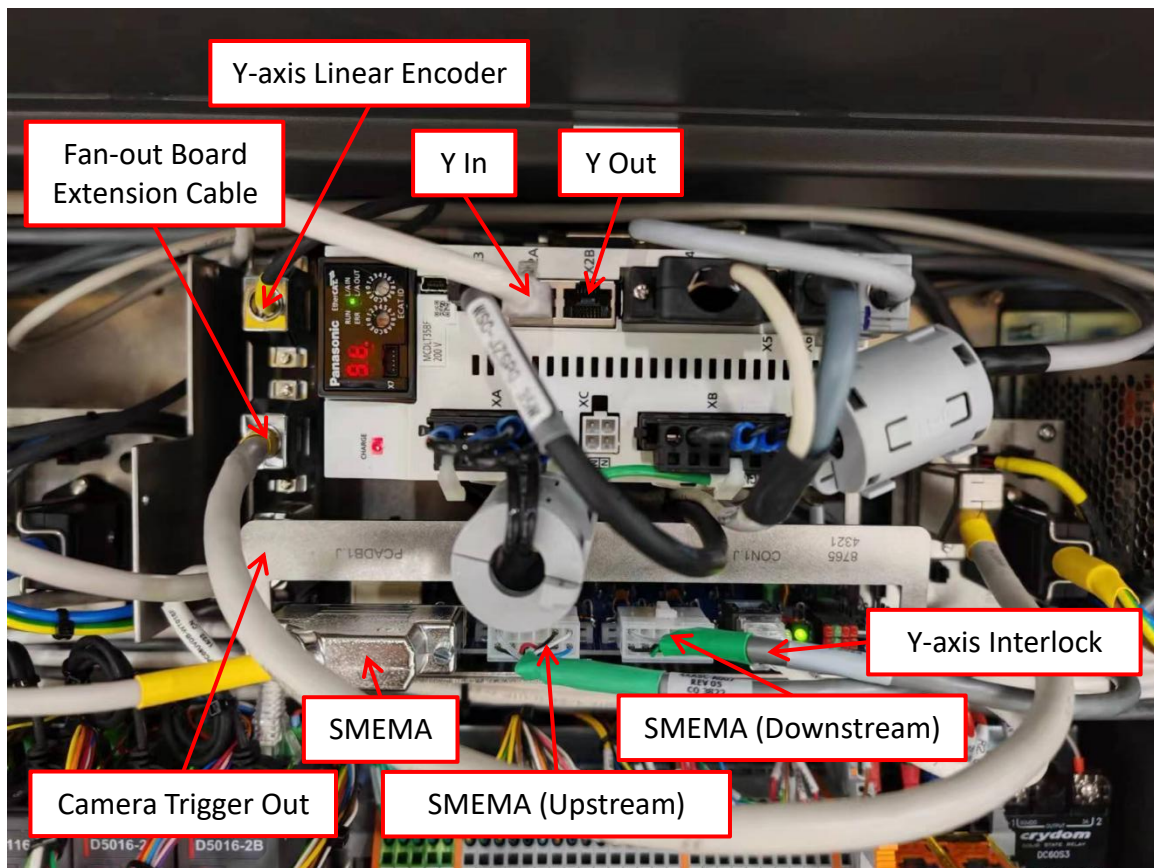


Figure 5 Y-axis Panasonic Ethercat Motion Driver Cable Labeling (1)

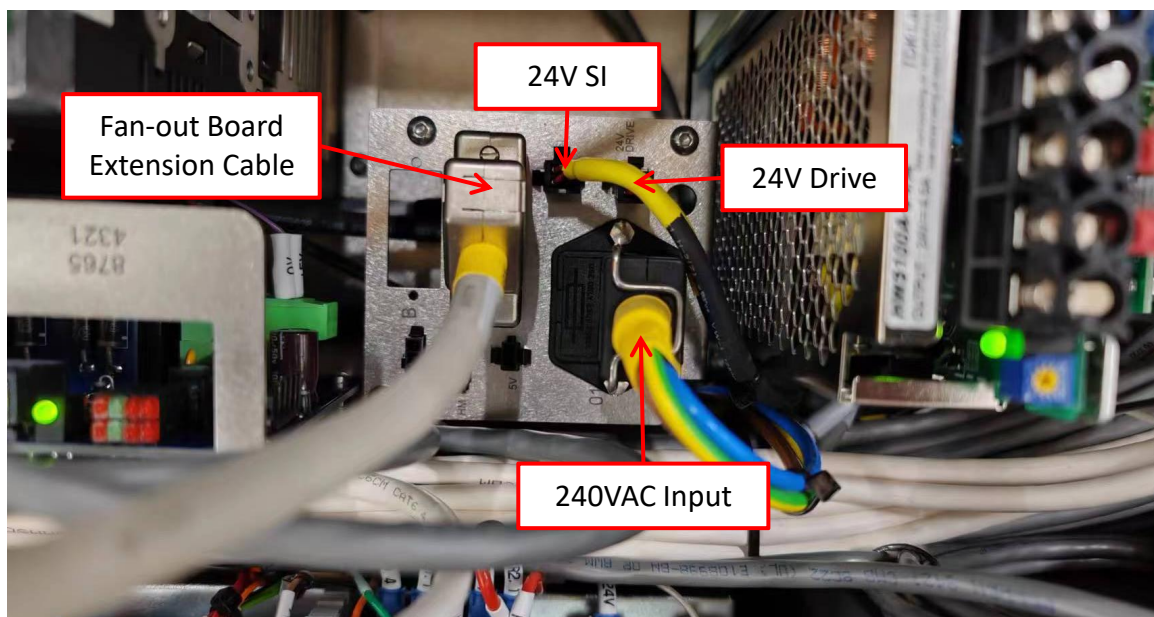


Figure 6 Y-axis Panasonic Ethercat Motion Driver Cable Labeling (2)

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4. After labeling all cable, disconnect the labeled cable from the motion driver.
5. Loosen 4x thumb screw at the base of the motion driver bracket.

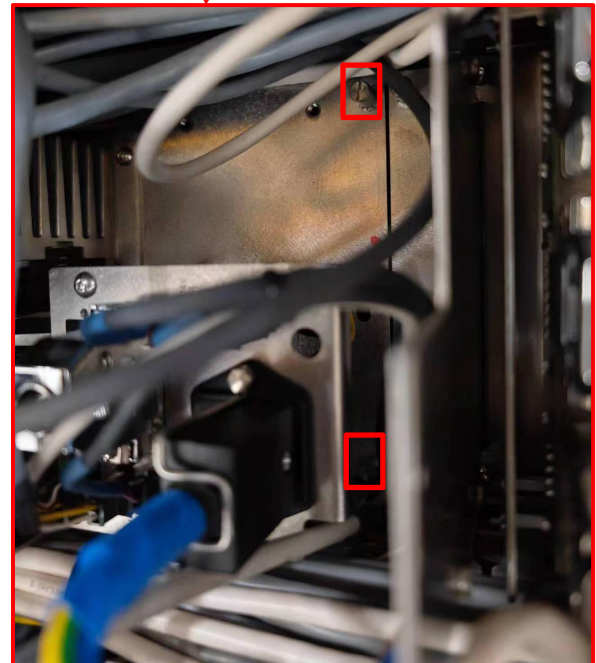
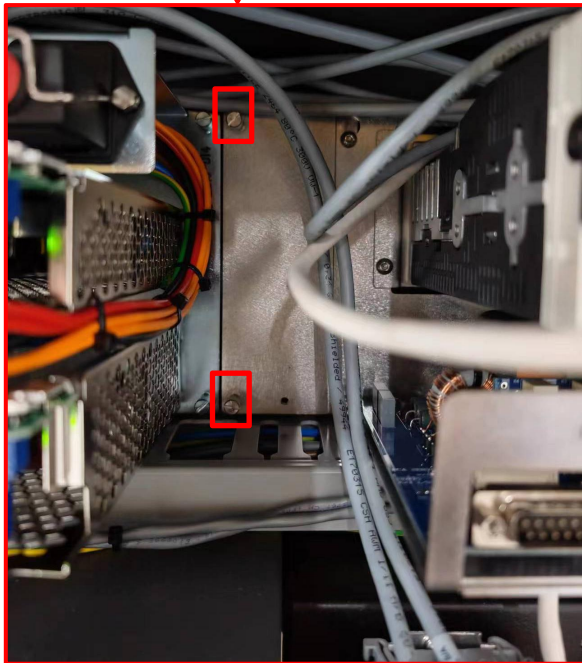
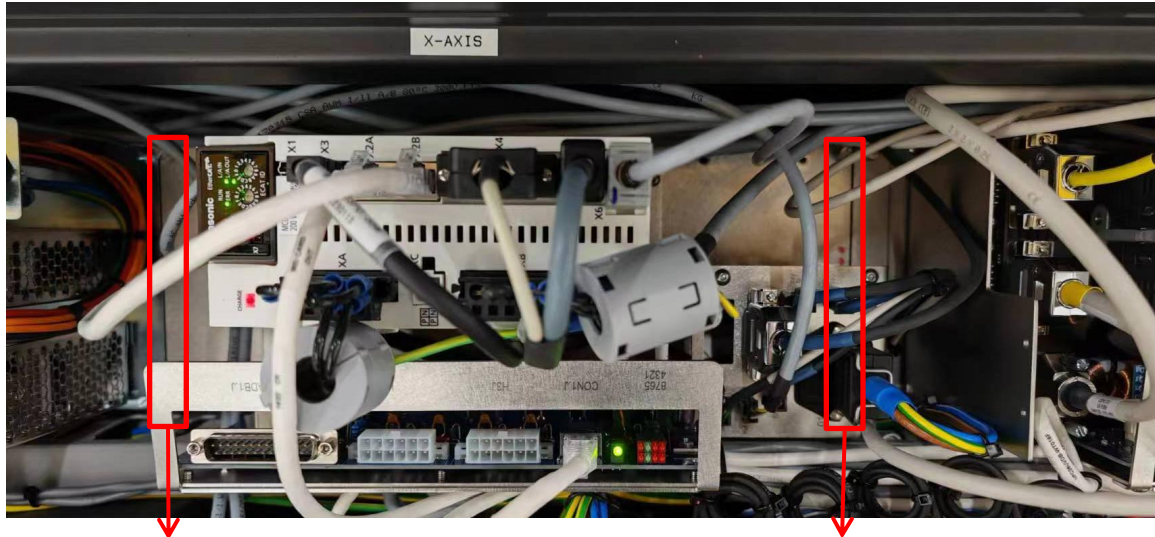


Figure 7 Thumb Screw Location

6. Remove the motion driver assembly from the SSCA.

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Ferrite Core and Fan-out Board Transfer

1. Prepare the new Panasonic Ethercat Motion Driver to be installed.
2. Transfer the ferrite core on the XA and XB connector from the old motion driver to the new motion driver.

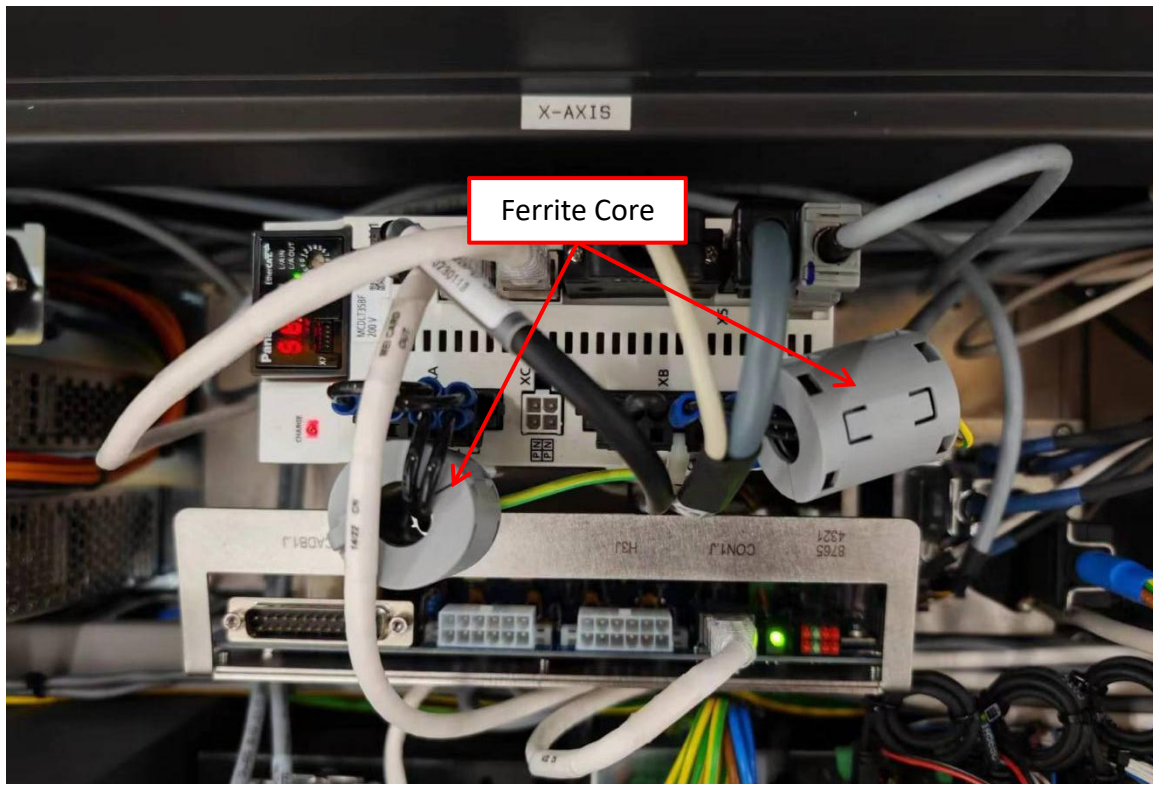


Figure 8 Ferrite Core

3. If the motion driver will be installed at Y-axis, transfer the fan-out board from the old motion driver to the new motion driver by removing two screws at the base of the fan-out board bracket.

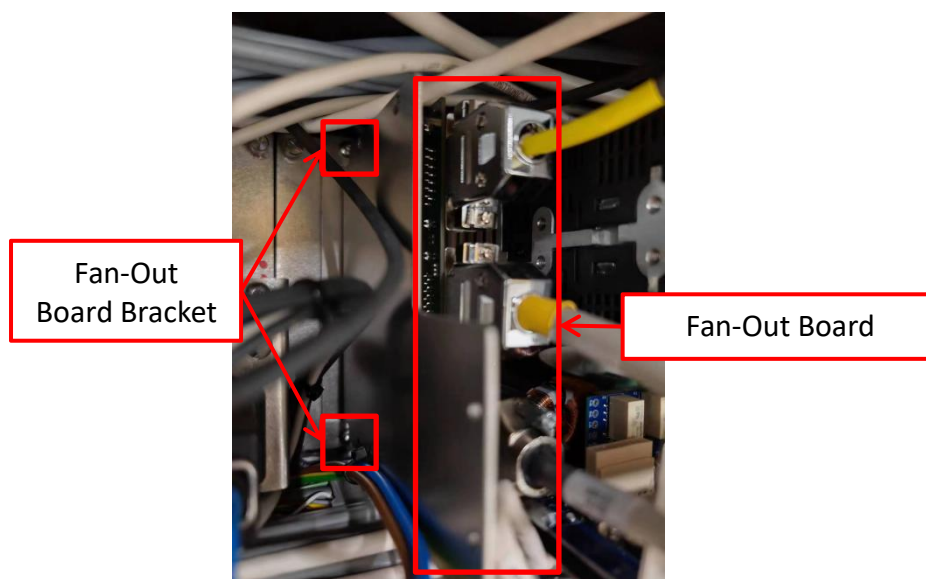


Figure 9 Fan-Out Board

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Panasonic Ethercat Motion Driver Assembly Installation

1. Place the Motion Driver Assembly into the SSCA according to the axis.

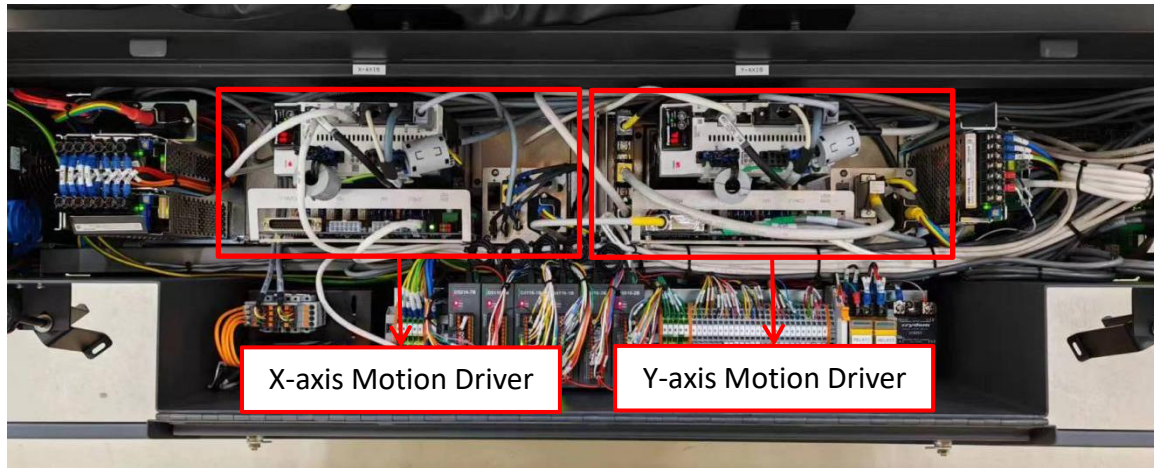


Figure 10 X & Y Axis Panasonic Ethercat Motion Driver

2. Tighten the 4 thumb screws at the base of the motion driver assembly bracket.

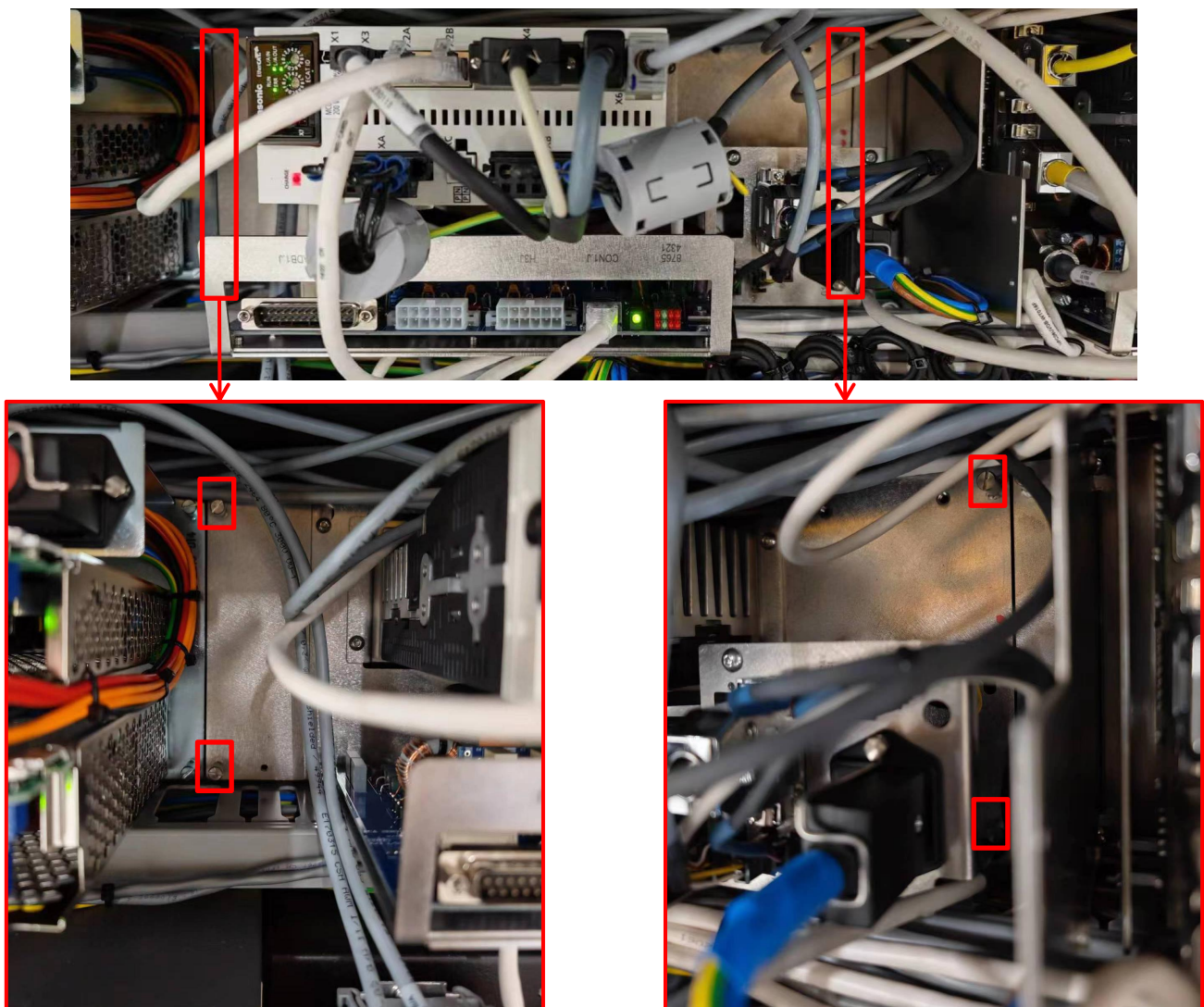


Figure 11 Thumb Screw Location

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3. Connect back the cables according to the label from the removal process.

X-axis Panasonic Ethercat Motion Driver

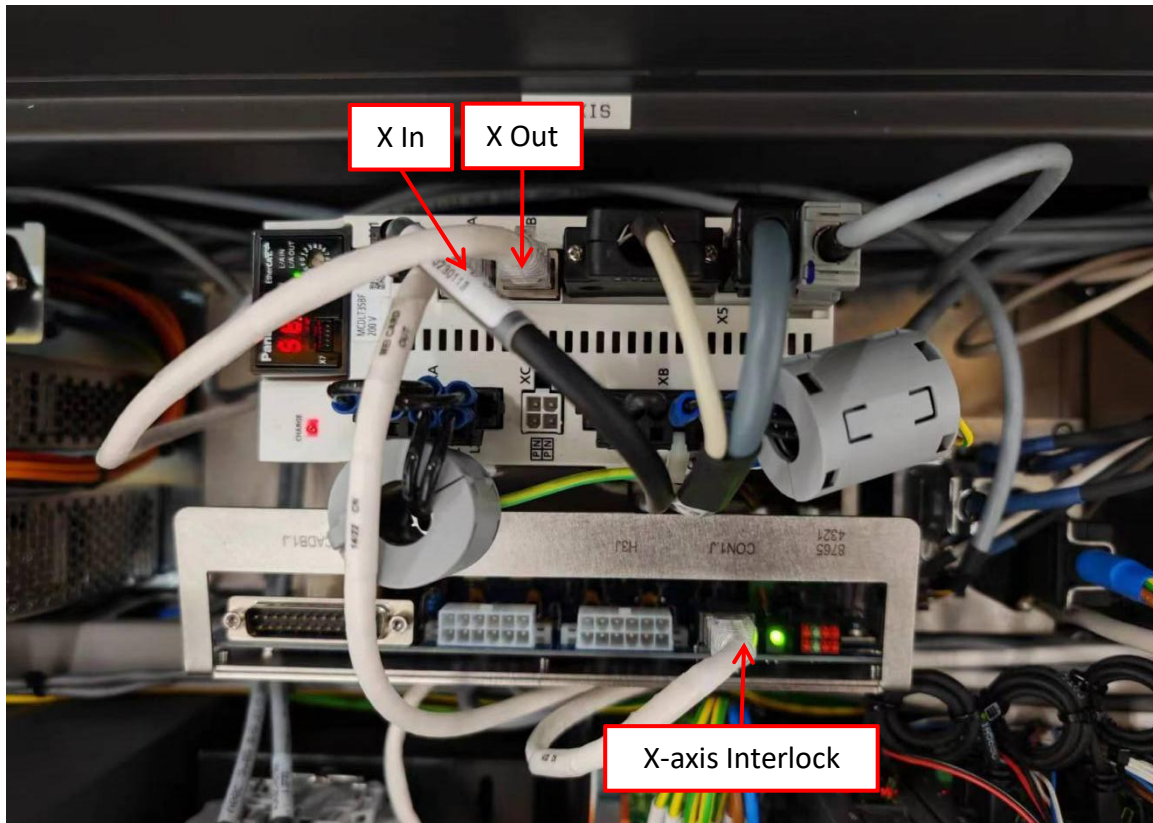


Figure 12 X-axis Panasonic Ethercat Motion Driver Cable Labeling (1)

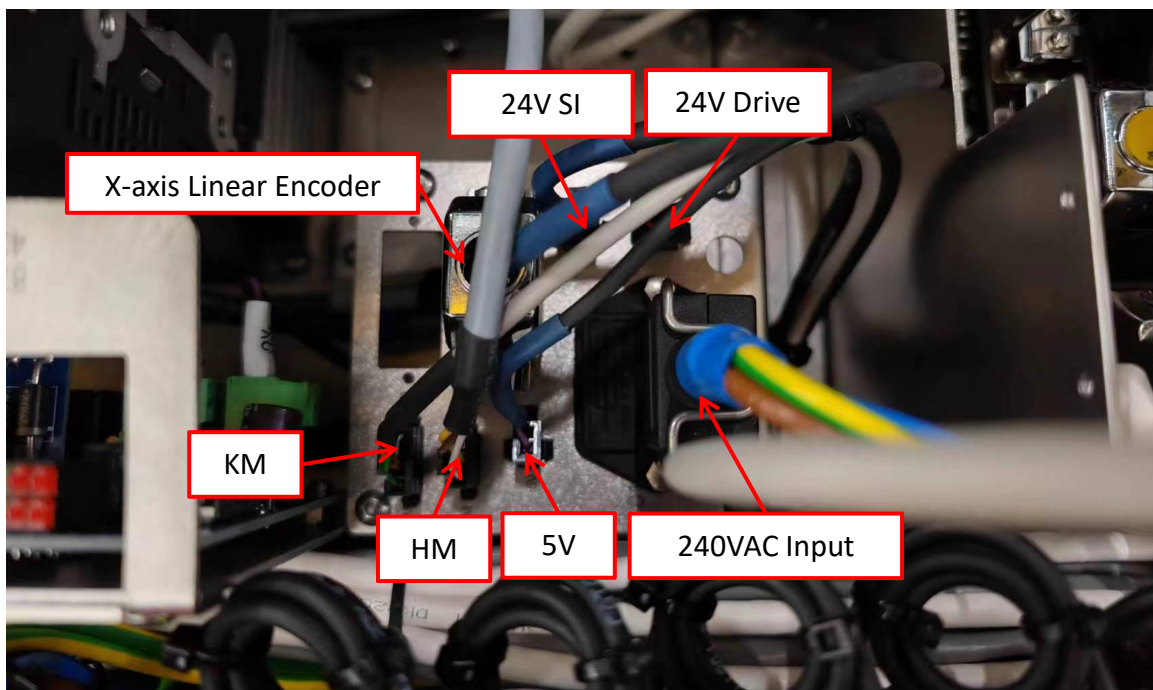


Figure 13 X-axis Panasonic Ethercat Motion Driver Cable Labeling (2)

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Y-axis Panasonic Ethercat Motion Driver

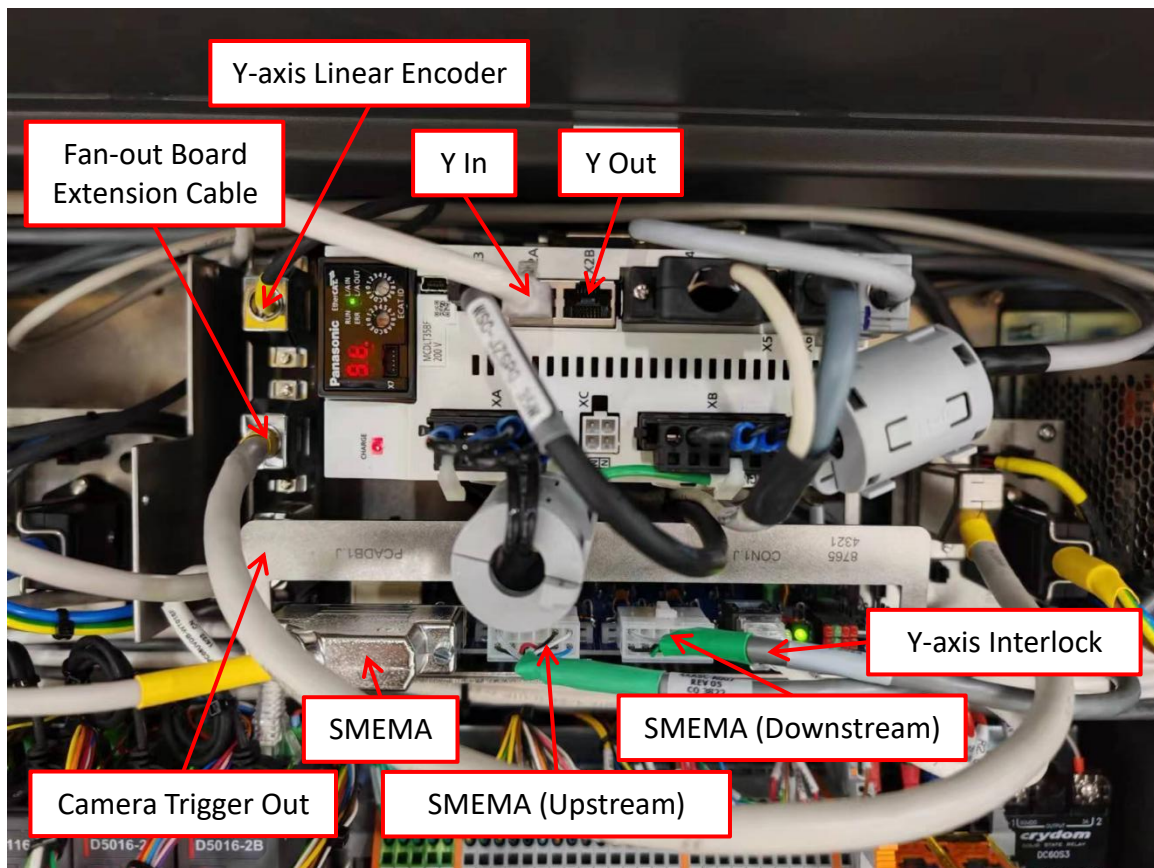


Figure 14 Y-axis Panasonic Ethercat Motion Driver Cable Labeling (1)

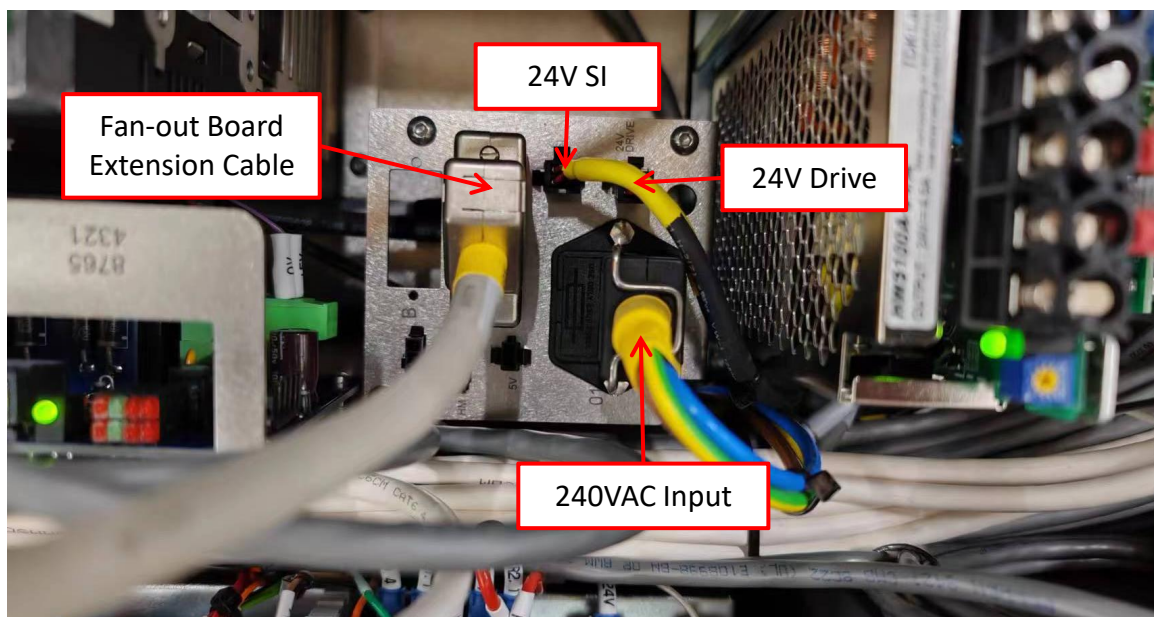


Figure 15 Y-axis Panasonic Ethercat Motion Driver Cable Labeling (2)

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Panasonic Ethercat Motion Driver Configuration

1. Prepare a USB Type A Male to Mini USB Type B Male cable (PN: 2530-0242).



Figure 16 USB Type A Male to Mini USB Type B Male cable

2. Connect Type A USB connector to Atom PC USB Port



Figure 17 Atom PC Connection

3. For X-axis configuration, connect Mini USB Type B connector to X-axis Motion Driver X1 port. For Y-axis configuration, connect Mini USB Type B connector to Y-axis Motion Driver X1 port.

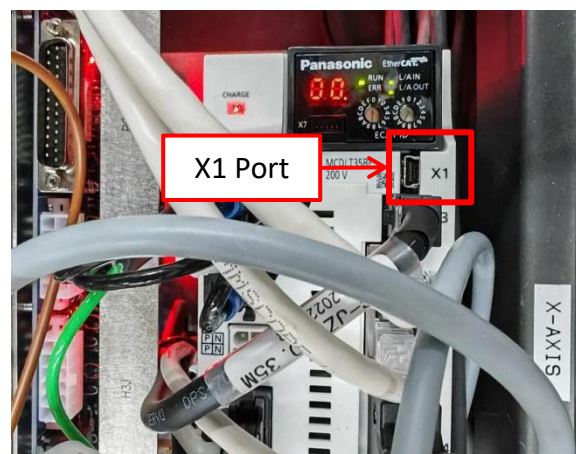


Figure 18 X1 Port

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4. Disconnect LAN Cable at ADVANTECH PCI-1203 Card.

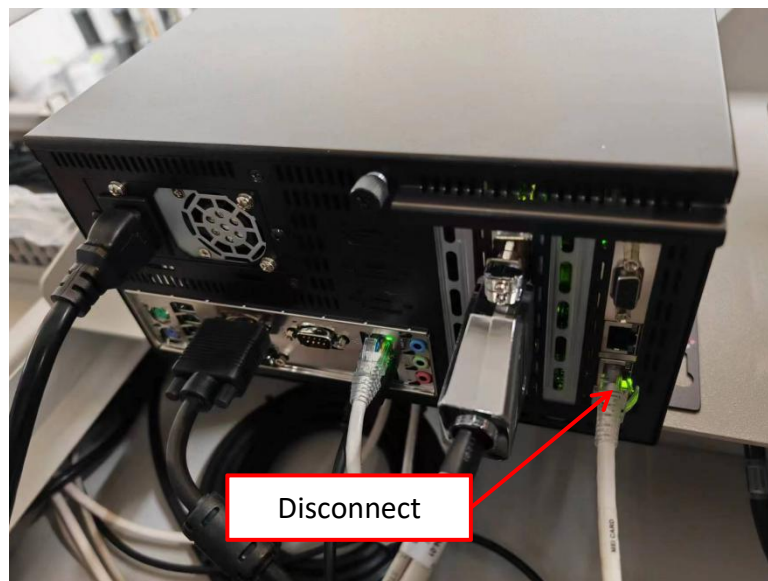


Figure 19 Disconnect ADVANTECH PCI-1203 LAN Cable

5. Switch On SSCA.

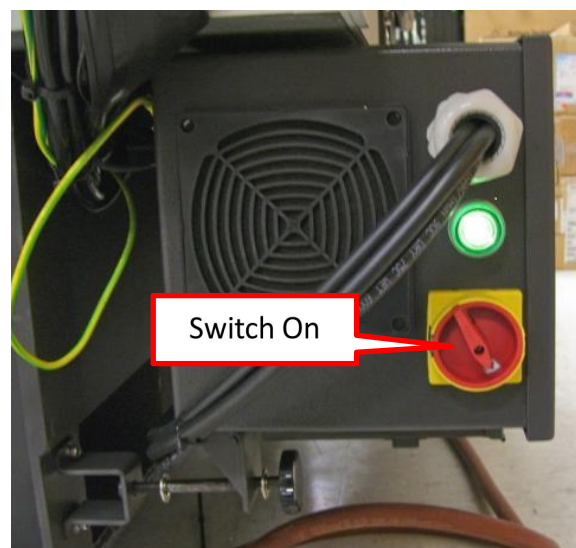


Figure 20 Turn Off SSCA

6. Remote into atom PC via 192.168.128.70.

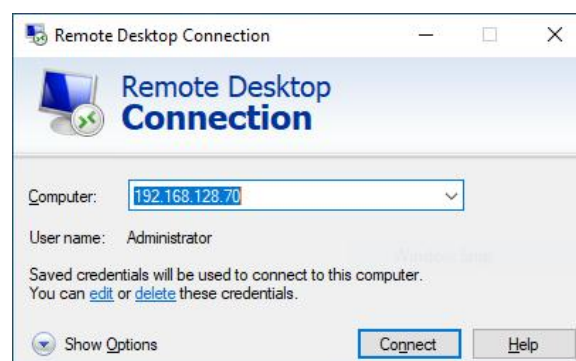


Figure 21 Remote Desktop Connection (192.168.128.70)

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7. Right click on “**PANATERM**” application shortcut at desktop and select “Run as Administrator”. If shortcut is unavailable, the application can be found at path:

“C:\Program Files (x86)\Panasonic Industry\MINAS”



Figure 22 PANATERM application shortcut

8. Select “Nickname Setting”.

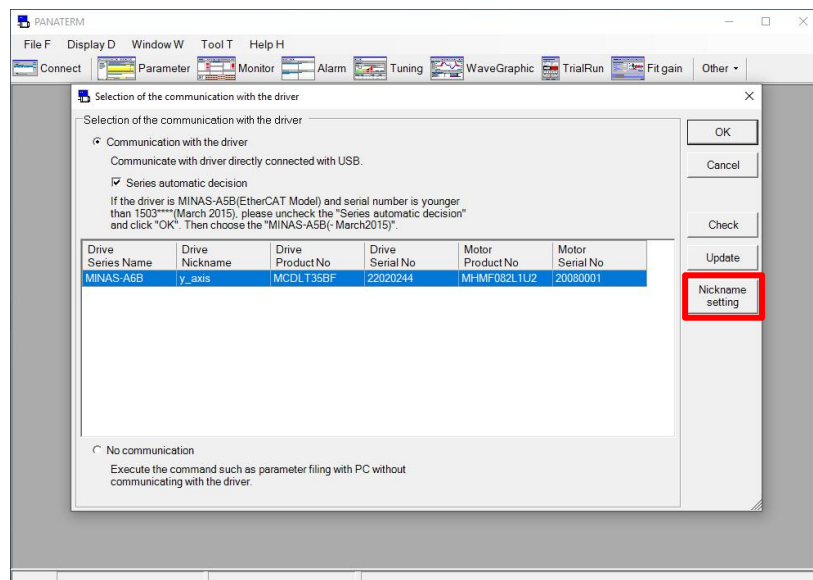


Figure 23 Nickname Setting

9. Assign nickname according to the axis by referring to Table 1.

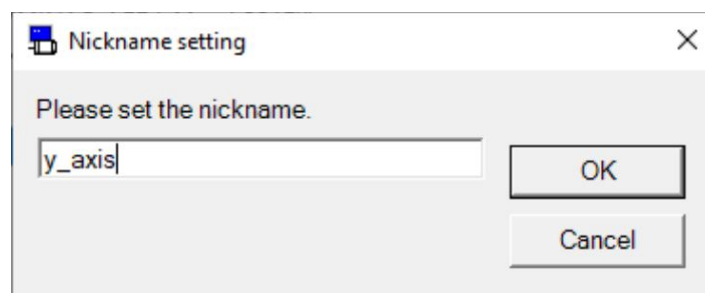


Figure 24 Assign Nickname

Axis	Nickname Setting
X-axis	x_axis
Y-axis	y_axis

Table 1 Nickname Setting of X and Y Axis

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10. Click “OK”.

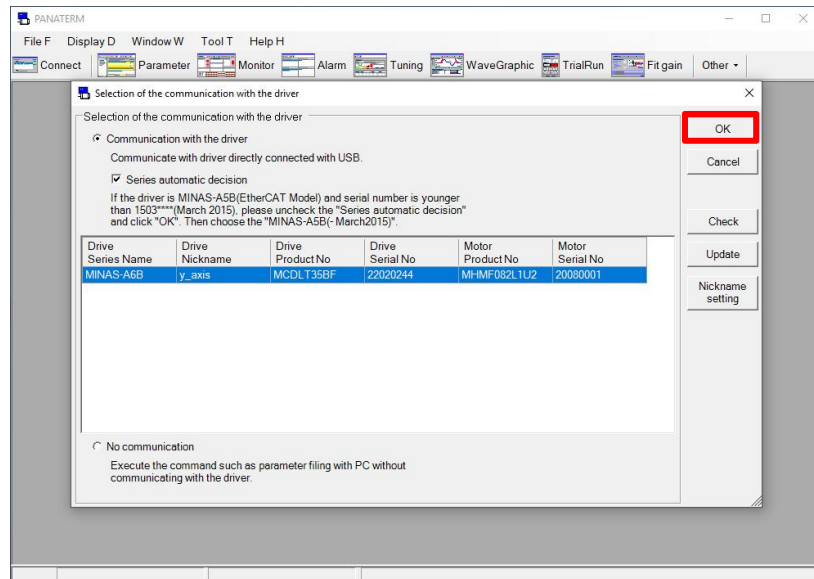


Figure 25 Driver Communication Selection

11. Go to “Display D” tab and select Object Editor.

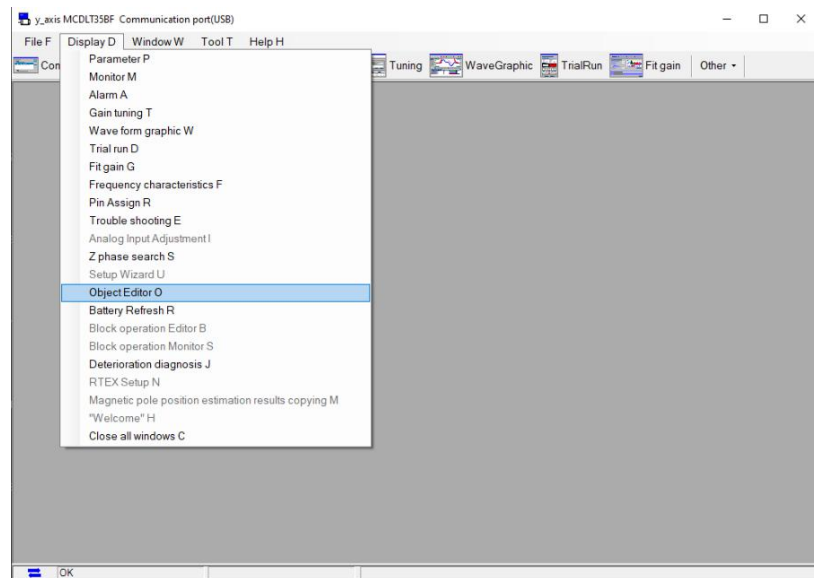


Figure 26 Object Editor

12. Select “Read from the file” and then click “OK”.

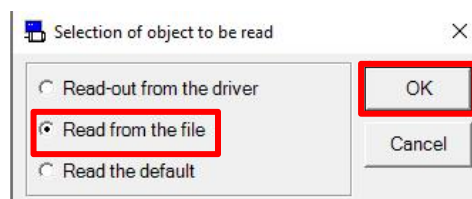


Figure 27 Selection of object to be read

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13. Go to below path and select the object file according to the axis by referring to Table 2. Click “OK” when done.

“C:\Program Files\AXI System\v810\5.0\s3Config\motionControl\motionDrive\ethercat\Panasonic_A6BF”

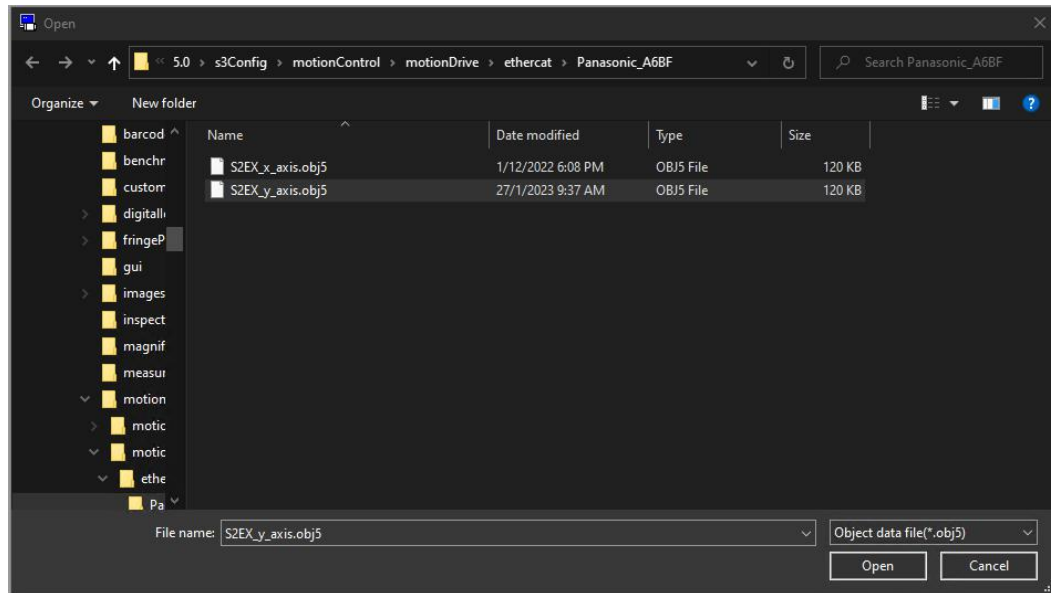


Figure 28 Object File

Axis	Object File
X-axis	S2EX_x_axis.obj5
Y-axis	S2EX_y_axis.obj5

Table 2 Object File

14. Click “OK”.

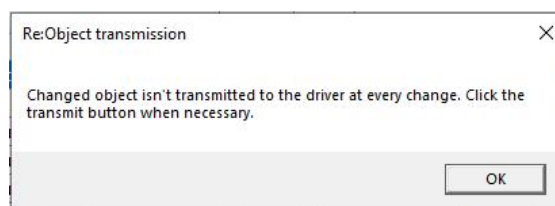


Table 3 Re: Object Transmission

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15. Click “Trans”.

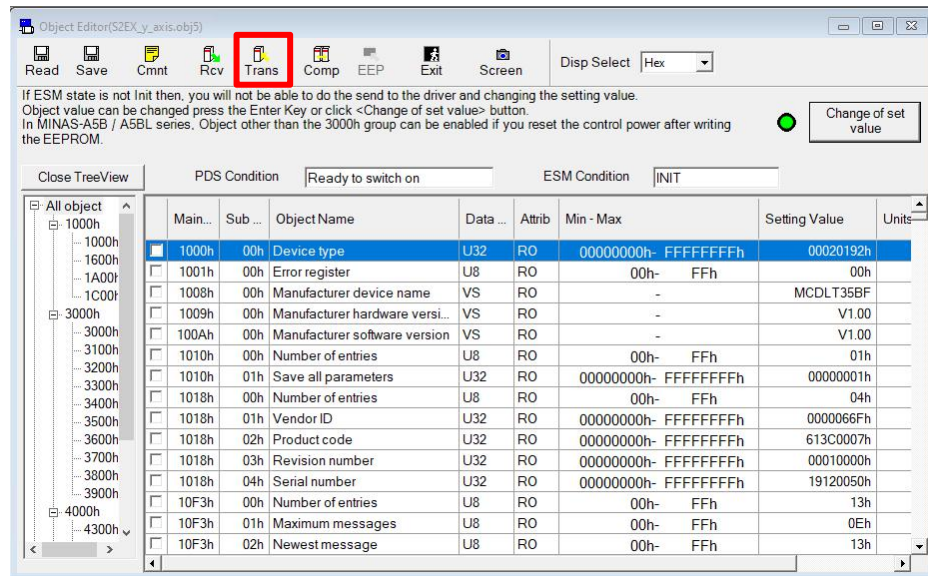


Figure 29 Object Editor (1)

16. Click “EEP”.

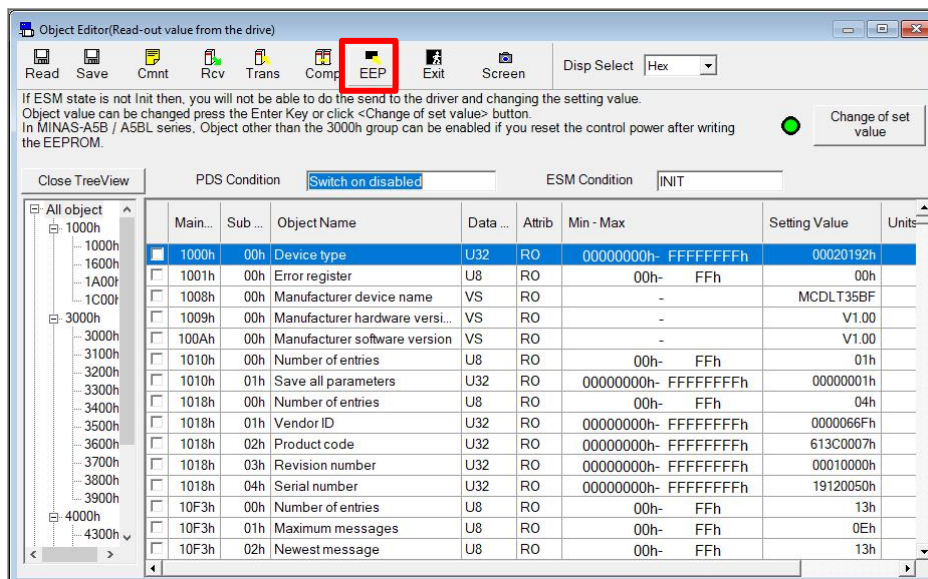


Figure 30 Object Editor (2)

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17. Click “OK”.

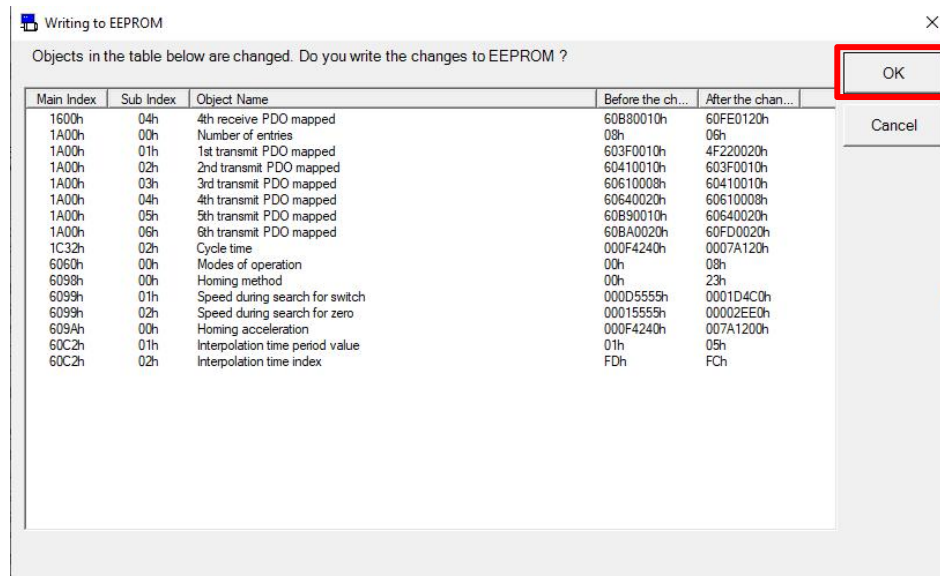


Figure 31 Writing to EEPROM

18. Wait until the process is complete.

19. Close the PANATERM application.

20. Connect back the LAN Cable at the ADVANTECH PCI-1203 Card.

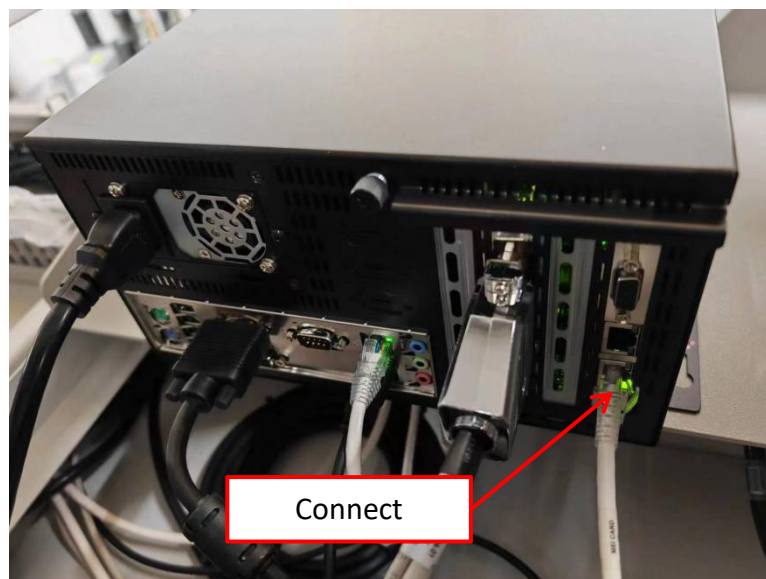


Figure 32 Connect ADVANTECH PCI-1203 LAN Cable

21. Power Cycle the SSCA.

22. Launch Common Motion Utility at path “C:\Program Files (x86)\Advantech\Common Motion\Utility”.



Figure 33 Common Motion Utility

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23. Go to My System > PCI-1203-06AE > Motion Ring and select the correct motion driver according to the node number by referring to Table 4.

Axis	Node
X-axis	0x001
Y-axis	0x002

Table 4 Axis Node Number

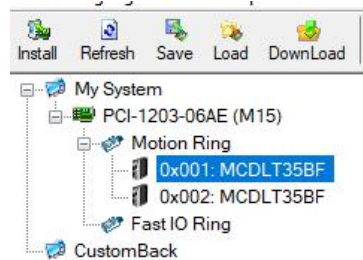


Figure 34 Node Selection

24. Go to “Information” tab.

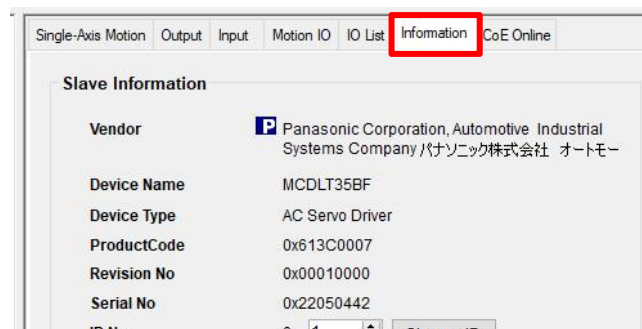


Figure 35 Information Tab

25. Modify the “ID no” according to the axis by referring to Table 5. Click “Change ID” when done.

Axis	ID No
X-axis	1
Y-axis	2

Table 5 Axis ID Number

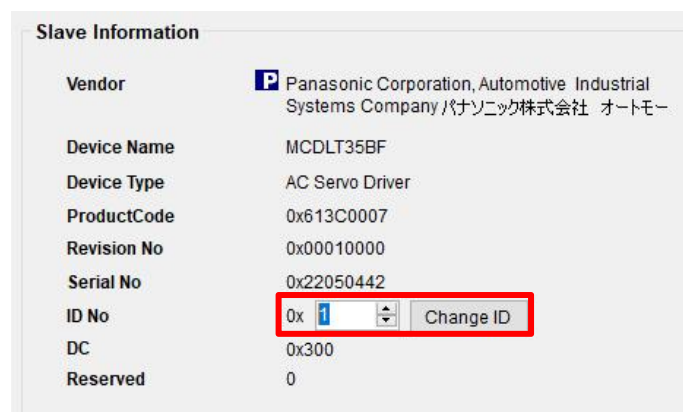


Figure 36 Assign ID No

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26. Click “OK” after successfully change ID No.

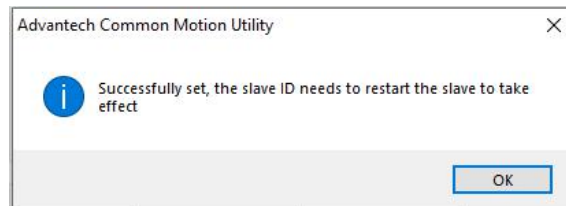


Figure 37 ID No Changed

27. Close the Common Motion Utility application.

28. Power cycle the SSCA for the changes to take effect.